Information System

Michael Cao

Topic 1: Logic Gate

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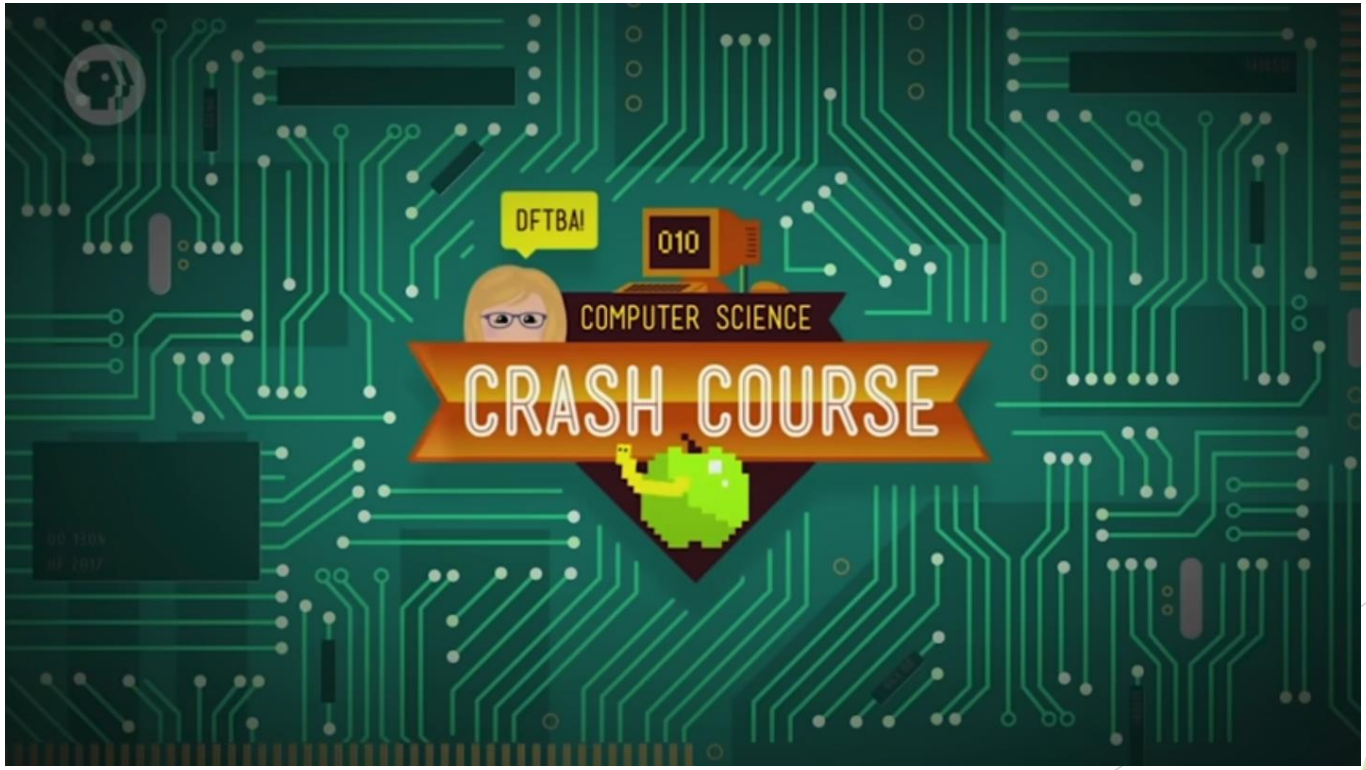
Learning Object

In this course, we will focus on the following concept:

- Basic logic gate concept
- Design of logic gate circuit

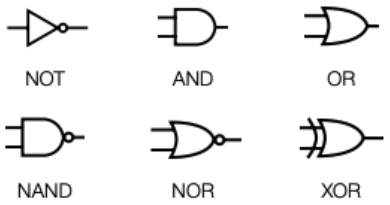
What is Logic Gate?

Logic gate is the foundation in computer for everything!



Syllabus of Logic Gate

3.2 Logic Gates and Logic Circuits

Candidates should be able to:	Notes and guidance
Use the following logic gate symbols: 	
Understand and define the functions of : NOT, AND, OR, NAND, NOR and XOR (EOR) gates	All gates except the NOT gate will have two inputs only.
Construct the truth table for each of the logic gates above	
Construct a logic circuit	From: • a problem statement • a logic expression • a truth table
Construct a truth table	From: • a problem statement • a logic circuit • a logic expression
Construct a logic expression	From: • a problem statement • a logic circuit • a truth table

10 Boolean logic

Candidates should be able to:	Notes and guidance
1 Identify and use the standard symbols for logic gates	• See section 4 for logic gate symbols
2 Define and understand the functions of the logic gates	• Including: – NOT – AND – OR – NAND – NOR – XOR (EOR) – the binary output produced from all the possible binary inputs • NOT is a single input gate • All other gates are limited to two inputs
3 (a) Use logic gates to create given logic circuits from a: (i) problem statement (ii) logic expression (iii) truth table	• Circuits must be drawn for the statement given, without simplification • Logic circuits will be limited to a maximum of three inputs and one output
(b) Complete a truth table from a: (i) problem statement (ii) logic expression (iii) logic circuit	• An example truth table with three inputs, for completion:
(c) Write a logic expression from a: (i) problem statement (ii) logic circuit (iii) truth table	

A	B	C	Output
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Points to be Stressed

Please try to answer these questions:

- What’s the truth table of XOR gate?
- What is ‘problem statement’?
- How to conduct a logic expression from a truth table?

Then finish the diagnostic test

Practical Usage of Logic Gate

Based on the video, please think of the questions:

- Why we use logic gate?
- What kind of problem can logic circuit solve?
- What will be the truth table of 1 bit adder? And logic circuit?
- What about considering the carry bit?
- What will be the design of 8-bit adder using logic circuit?
- Any other function of circuit you can build? Use online simulator !

Topic 2: Number System

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Learning Object

In this course, we will focus on the following concept:

- Number system
- Two’s complement
- BCD code
- ASCII code and Unicode

Number System

Binary number is the foundation in computer science:

- Binary is used because switches only have ‘on’ and ‘off’ states
- Hexadecimal is used because it’s easier for human to read
- The key of transformation is to focus on the weight
- Special attention for the definition of ‘prefix’

Activity: explore the definition of positional and un-positional number system

Decimal prefix name	Symbol used	Factor applied to the value
kilo	k	10^3
mega	M	10^6
giga	G	10^9
tera	T	10^{12}

Binary prefix name	Symbol used	Factor applied to the value
kibi	Ki	2^{10}
mebi	Mi	2^{20}
gibi	Gi	2^{30}
tebi	Ti	2^{40}

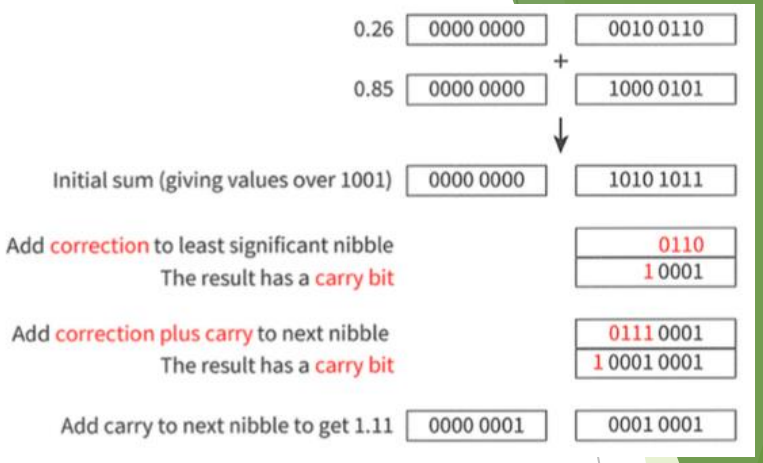
Two’s Complement

We use two’s complement for calculation:

- If we use original number, there will cause error
- Multiple ways to calculate the two’s complement
- The philosophy is to find a number that have the same result with carry or overflow
- -128 – 127 is the range of two’s complement
- Explore how to transfer a negative two’s complement to denary number directly

Practice for deriving the two’s complement, as well as addition calculation

BCD Code



BCD stands for Binary Coded Decimal:

- Useful in applications that require single denary digits be used
- BCD represents one denary digit in a nibble
- If over than 9, then plus 6 (0110)

Challenge: Use python to create a BCD adder, the input is 2 positive number, convert each digit to binary, use BCD rules to add, then output denary result

ASCII and Unicode

We need a coding scheme to store text in a computer:

- ASCII: American Standard Code for Information Interchange
  - No need to remember each code, but know the rules
  - Limited characters with 1 byte length (Extended)
- Unicode: Cover more characters with 2 bytes

Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
32	20	<SPACE>	64	40	@	96	60	`
33	21	!	65	41	A	97	61	a
34	22	“	66	42	B	98	62	b
35	23	#	67	43	C	99	63	c
36	24	$	68	44	D	100	64	d
37	25	%	69	45	E	101	65	e
38	26	&	70	46	F	102	66	f
39	27	‘	71	47	G	103	67	g
40	28	(	72	48	H	104	68	h
41	29	)	73	49	I	105	69	i
42	2A	*	74	4A	J	106	6A	j
43	2B	+	75	4B	K	107	6B	k
44	2C	,	76	4C	L	108	6C	l
45	2D	-	77	4D	M	109	6D	m
46	2E	.	78	4E	N	110	6E	n
47	2F	/	79	4F	O	111	6F	o
48	30	0	80	50	P	112	70	p
49	31	1	81	51	Q	113	71	q
50	32	2	82	52	R	114	72	r
51	33	3	83	53	S	115	73	s
52	34	4	84	54	T	116	74	t
53	35	5	85	55	U	117	75	u
54	36	6	86	56	V	118	76	v
55	37	7	87	57	W	119	77	w
56	38	8	88	58	X	120	78	x
57	39	9	89	59	Y	121	79	y
58	3A	:	90	5A	Z	122	7A	z
59	3B	;	91	5B	[	123	7B	{
60	3C	<	92	5C	\	124	7C	|
61	3D	=	93	5D	]	125	7D	}
62	3E	>	94	5E	^	126	7E	~
63	3F	?	95	5F	_	127	7F	<DELETE>

Topic 3: Media and Compression

Michael Cao

Learning Object

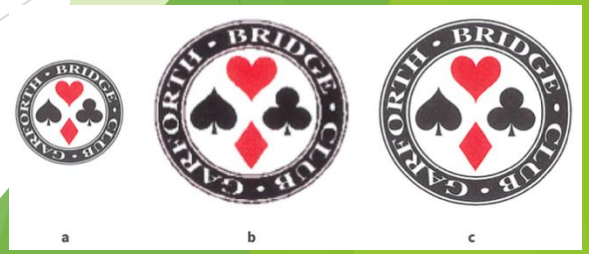
In this course, we will focus on the following concept:

- Bitmap and vector graphic
- Sound
- Compression technique

Bitmap

Bitmap is generally used to store an image:

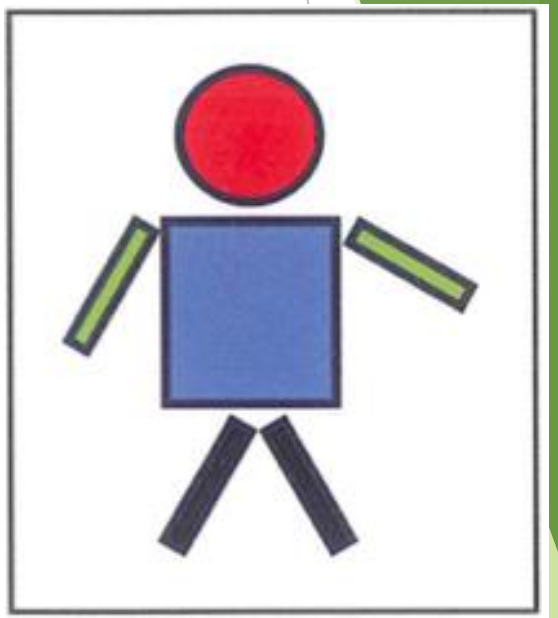
- Pixel: the smallest component of bitmap. (essential information?)
- Color depth: the number of bits used to represent one pixel
- Bit depth: the number of bits used to represent each channel
- Resolution: image resolution and screen resolution
- File Header: a set of bytes at the beginning identifies the files and contains information about the coding used



Vector Graphic

Vector graphic is usually created by human:

- Drawing Object: a component defined by geometric formulae and associated properties
- Drawing List: the list of shapes involved in an image // a list that stores the command/description required to draw each object
- Property: Define one aspect of the appearance of the drawing object



Print | Code History

```
# Fill me in!
Rect(0,0,400,400,fill='papayaWhip')
#eyebrows
Circle(160,200,120,fill=None,border='black',borderWidth=4)
Circle(240,200,120,fill=None,border='black',borderWidth=4)
Rect(40,200,400,200,fill='papayaWhip')
Rect(160,80,80,120,fill='papayaWhip')
Rect(120,100,160,100,fill='papayaWhip')
#Eyes
Circle(120,200,20)
Circle(280,200,20)
Oval(120,190,20,10,fill='white')
Oval(280,190,20,10,fill='white')
Oval(120,210,20,10,fill='white')
Oval(280,210,20,10,fill='white')
Rect(110,205,20,5)
Rect(270,205,20,5)
#mouth
Oval(200,295,80,120,fill=None,border='black',borderWidth=4)
Rect(160,200,80,120,fill='papayawhip')
Rect(200,280,40,120,fill='papayawhip')
Rect(170,370,100,30,fill='papayawhip')
Circle(170,280,40,fill=None,border='black',borderWidth=4)
Circle(230,280,40,fill=None,border='black',borderWidth=4)
Rect(120,240,160,40,fill='papayawhip')
Rect(180,280,40,20,fill='papayawhip')
Circle(200,300,6)
Circle(165,320,3)
```

Activity

Please make a comparison between vector graphic and bitmap from:

- Where it come from? Easy to create?
- What’s the basic component?
- What’s the feature?
- Occupation of space?
- Realistic? Vivid?
- The best use situation?
- ETC.

Sound

Sound is a continuous format of media:

- **ADC and DAC**: convert between analogue and digital data
- **Sampling**: taking measurements at regular intervals and storing the values
- **Sampling resolution**: the number of bits used to store each sample
- **Sampling rate**: the number of samples taken per second

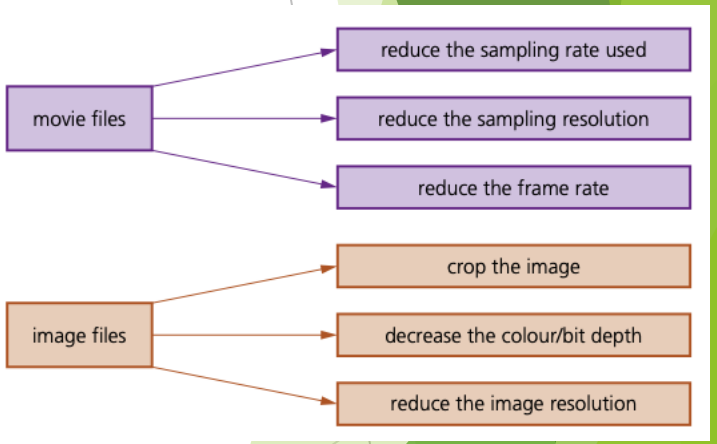
Calculate the file size of images, sound files and videos

Compression

Compression is useful to reduce file size:

- File Size: Calculated by unit and quantity. (also the logic of compression)
- Lossless compression: original file can be restored
  - Coding map table (use code.org to explore)
  - Run-length encoding (consecutive repeating patterns)
- Lossy compression: some data is lost permanently
  - Perceptual shaping
  - Crop or reduce resolution of depth
  - Relative mapping (adjacent frames in video)

Challenge: Write python code to implement RLE

Presentation: design / introduce a compression method by your own

Hint: Introduction can include from thesis or existed compression format